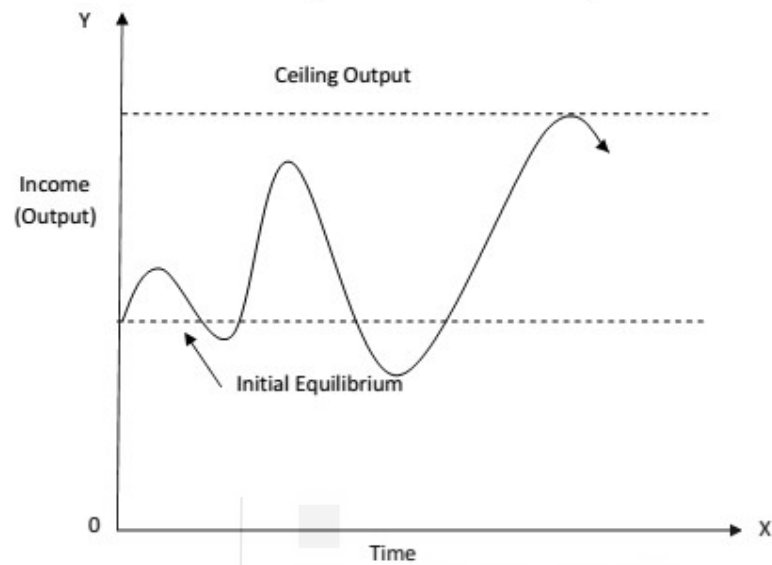
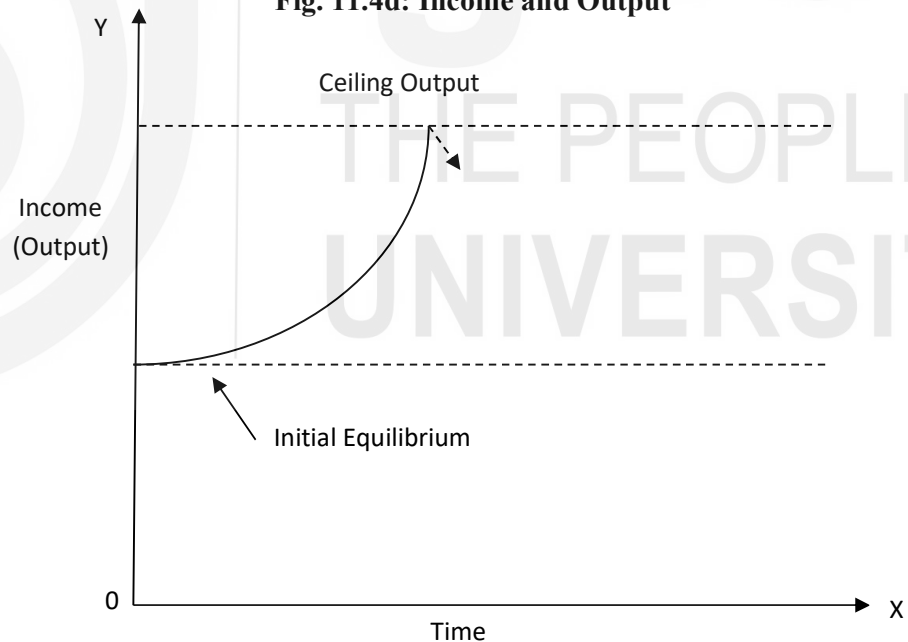


Fig. 11.4c: Income and Output



The region D provides the combinations of c and v which cause income to move upward or downward at an increasing rate. This is depicted in Fig. 11.4d.

Fig. 11.4d: Income and Output



In a special case when values of c and v lie in the region E, they produce fluctuations in income of constant amplitude and are shown in Fig. 11.4e. It is worth noting that all the above five cases do not give rise to cyclical fluctuations or business cycles. It is only combinations of c and v lying in the regions B, C and E that produce business cycles.

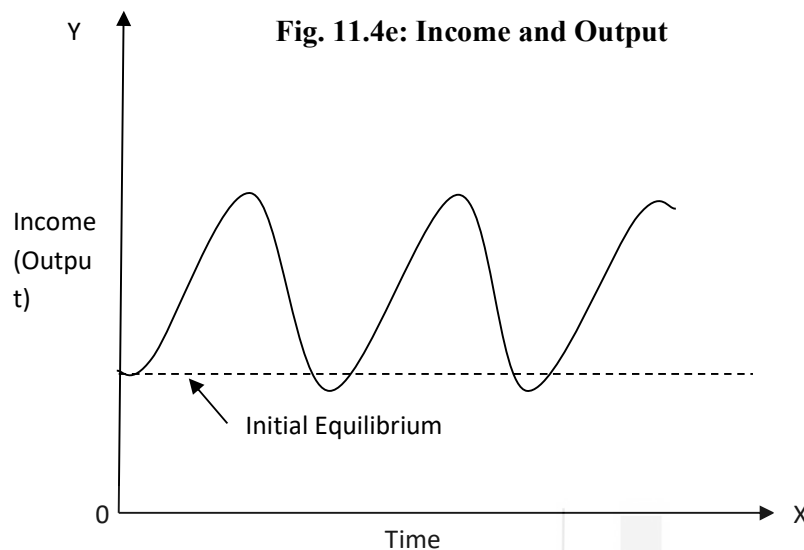


Fig. 11. 4 Panel (a) to (e) shows different patterns of income (output) movements for various values of c and v which respectively determine the magnitudes of multiplier and accelerator.

11.4.3 Political Business Cycles

The term *political business cycle* is used mainly to describe the stimulation of the economy just prior to an election in order to improve prospects of the incumbent government getting re-elected. Expansionary monetary and fiscal policies have politically popular consequences in the short run, such as falling unemployment, economic growth, and benefits from government spending on public services. However, the same policies, especially if pursued to excess, are found to have unpleasant consequences in the long term, such as accelerating inflation and damaging the foreign trade balance. Thus, they can harm the long-term growth potential of the economy. Governments prefer to pursue popular expansionary monetary and fiscal policies for a short horizon immediately before an election. However, being aware of its adverse effects, they do not continue with those measures after they get elected. Thus, after the election is over, government often reverses earlier decisions, which may include cutting spending, slowing the growth of money supply, and allowing interest rates to rise. As a result, the regular holding of elections will produce cyclical fluctuation of economic activity because of recurring patterns of government stimulus and restraint in order to induce an artificial boom in the election time.

11.5 DATING OF BUSINESS CYCLES

Neutralizing the effect of business cycles, particularly the downslides, is a key objective of macroeconomic policy. Understanding recessions and expansions of business cycles is essential for designing of proper frameworks for stabilisation. This requires identifying the chronology of business cycle turning points. In the

US, the National Bureau of Economic Research (NBER) has a dedicated research program for identifying the dates of business cycle turning points. Similarly the Centre for Economic Policy Research (CEPR) Euro Area Business Cycle Dating Committee establishes the chronology of recessions and expansions of Euro Area member countries. In India there have been some attempts to determine the chronology of business cycles.

The NBER's Business Cycle Dating Committee maintains a chronology of the US business cycle. The chronology comprises alternating dates of peaks and troughs in economic activity. A recession is a period between a peak and a trough, and an expansion is a period between a trough and a peak. The NBER does not define a recession in terms of two consecutive quarters of decline in real GDP. Rather, a recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real GDP, real income, employment, industrial production, and wholesale-retail sales. Similarly, during an expansion, economic activity rises substantially, spreads across the economy, and usually lasts for several years. Thus, the NBER approach identified cycles as recurrent sequences of alternating phases of expansion and contraction in the levels of a large number of economic time series. This working definition, in substantially its present form, has been in use at the NBER for over fifty years and is currently employed by the NBER to identify and date the US business cycle. These dates are widely accepted by government, academic, and business analysts. The definition served as the basis for the NBER's pre-World War II business cycle chronologies for England, France, and Germany as well as the US.

In both recessions and expansions, brief reversals in economic activity may occur—a recession may include a short period of expansion followed by further decline; an expansion may include a short period of contraction followed by further growth. The Committee applies its judgment based on the above definitions of recessions and expansions and has no fixed rule to determine these upturns and downturns.

The Committee does not have a fixed definition of economic activity. It examines and compares the behaviour of various measures of broad activity: real GDP measured on the product and income sides, economy-wide employment, and real income. The Committee also may consider indicators that do not cover the entire economy, such as real sales and the Federal Reserve's index of industrial production (IIP).

11.6 ECONOMIC INDICATORS

The objectives of macroeconomic policy have long included the avoidance of protracted recessions in which resources go underutilized, and of periods of unsustainable growth that jeopardize reasonable price stability. An important part of the job of the central bank is therefore to gather information on the current state of the economy and if possible, forecast future economic conditions. The

theoretical concept of measuring current business activities using economic series such as GDP, sales, investment, stock prices, is rather simple though its practical application is difficult. Usually, the time pattern of these fluctuating economic series is diverse. While some economic series are expanding at a given point in time, others have already reached their upper turning point and still others are on the downswing; a few economic activities might even be at a lower turning point. Thus the issue arises is how to measure the aggregate state of the economy using these various individual economic series as they have diverse timing patterns.

Economic indicators were conceived at the NBER originally by Mitchell and Burns. This approach requires monitoring of economic variables that tend to be sensitive to cyclical changes no matter what their cause. Burn and Mitchell studied a group of about 487 economic variables to see if turning points in the variables persistently led, coincided with, or lagged turning points in the US business cycle. Seventy one series were chosen and ordered according to the average lead or lag with regard to the reference revivals. For example, six time series had no average lead or lag. On the average, the leading series were from one to ten months ahead of the reference revivals. The lagging series were on the average from one to twelve months behind. The idea of leading, lagging, and coinciding (reference) series was developed even further by G.H. Moore in 1950. Out of 801 series, he chose 21 series which passed a test for conformity and timing; conformity meaning the consistency with which the series have conformed to business cycles and timing referring to the consistency with which a time series' turning point has led, lagged or roughly coincided with the reference data. Those series were selected which were similar in timing at peaks and troughs and were classified into three groups: leading, roughly coincident, and lagging.

11.6.1 Leading Indicators

Leading economic indicators help us in assessing where the economy is heading. They indicate what is coming, such as moving from a peak phase into a contraction, before it actually happens.

One of the most significant leading indicators is the stock market itself, gauged by an index such as the S&P 500. It will begin to rise before economic conditions seems favourable, and it will be begin to decline before economic conditions seem to warrant it. Another important leading indicator is interest rates. Low interests stimulate borrowing and buying, which favours the economy. An increase in interest rates shows the economy is doing well, but eventually rising interest rates lead to a slow down because less people borrow money to start new projects.

11.6.2 Lagging Indicators

Unlike leading indicators, lagging indicators shift after the economy changes. Although they do not typically tell us where the economy is headed, they indicate how the economy changes over time and can help to identify long-term trends. Lagging economic indicators reveal past information about the economy.

The GDP is how much a country is producing. There is significant lag time between when the data is compiled and when it is released, yet it is still an important indicator. Many consider a recession to be underway if two quarters see back-to-back declining GDP. Other indicators, such as the Consumer Price Index (CPI), are also sometimes considered lagging indicators, since they reveal information that is already known to most consumers.

11.6.3 Coincident Indicators

Coincident indicators (more or less) in parallel with the general economic conditions and therefore reflect the current status of the economy. They give consumers, business leaders, and policy makers an idea about where the economy is currently, right now. When the economy rises today, then coincident indicators are also rising right now. And when the economy declines today, then coincident indicators are also declining right now. Typical examples of coincident indicators are industrial production or turnover.

The three types of indicators i.e., leading, lagging and coincident indicators under five categories namely, Investment, Consumption, Employment, Prices and Money are presented in Table 11.1.

Table 11.1: Indicators of Business Cycle		
Leading	Roughly Coincident	Lagging
<i>I. Investment in Fixed Capital and Inventories</i>		
New building permits; housing starts; residential fixed investment New business formation New capital appropriations; contracts and orders for plant and equipment Change in business inventories	Production of business Equipment; Machinery and equipment sales	Backlog of capital Appropriations; Business expenditures for new plant and equipment; and trade inventories

<i>II. Consumption, Trade, Orders, and Deliveries</i>		
New orders for consumer goods and materials; Change in unfilled orders, durable Goods; Vendor performance (speed of deliveries); Index of consumer sentiment	Production of consumer Goods; and trade sales	
<i>III. Employment, Production, and Income</i>		
Average workweek; overtime Hours; Accession rate; layoff rate; New unemployment insurance Claims; Productivity (output per hour); Rate of capacity utilization	Non-agricultural employment; unemployment rate; GNP; personal income; industrial production	Average duration of unemployment; long term unemployment
<i>IV. Prices, Costs, and Profits</i>		
Bond prices; Stock prices; Sensitive materials prices; Profit margins; Total corporate profits; net cash flows		Unit labour costs; Labour share in national income
<i>V. Money, Credit, and Interest</i>		
Monetary growth rates; Change in liquid assets; Change in consumer credit; Total private borrowing; Real money supply	Velocity of money	Short-term interest rates; Bond yields; Consumer credit outstanding; Commercial and industrial loans outstanding

Note: The selection is based on the US indicators published in *Business Conditions Digest*, a monthly report by the Bureau of Economic Analysis, US Department of Commerce

11.7 EMPIRICAL ANALYSIS

Researchers have faced a many of challenges while studying business cycles. One of the major challenges is the problem of representing complex features of economic data. Kuznets, Mitchell, Burns and many others employed various techniques (i.e., moving averages, piecewise trends, etc.) to study the fluctuations of variables around the long term path. As you know, there are various techniques of data analysis. However, there is no single correct way to represent these components.

During the nineteenth century researchers focused on real variables such as agricultural output to explain business cycle. The classical economists assumed perfectly competitive markets and flexibility in prices and wage rate; thus ruled out the possibility of business cycle. According to them, money was neutral in its effects. Later on, during the twentieth century, economists ascribed monetary variables such as money supply, interest rates and prices important variables causing business cycles.

D. K. Backus and P. J. Kehoe in their study titled, “International Evidence on the Historical Properties of Business Cycles” in 1992 took a group of macro variables, in addition to the GDP, for analysis of business cycle. They examined the cyclical behaviour of private consumption expenditure, fixed investment, government purchases and net exports in ten countries, namely, Australia, Canada, Denmark, Germany, Italy, Japan, Norway, Sweden, the United Kingdom, and the US. They find a great deal of regularity in the cyclical behaviour of real quantities. The patterns of variation in all the variables, however, are not similar. The magnitude of output fluctuations varies across countries and over time. Further, the relationships among macroeconomic variables are remarkably stable. For example, investment is about 2 to 4 times more volatile than output while the variation in consumption is almost the same as the variation in output. Both investment and consumption are strongly procyclical. Trade balance is generally countercyclical, exhibiting larger deficits during booms than during recessions. The exception to this regularity in quantities is government purchases, which exhibit no systematic cyclical tendency.

I. Konstantakopoulou and E. G. Tsionas in their work on “Half a century of empirical evidence of Business Cycles in OECD countries” in 2014 examine the existence of common business cycles in OECD countries using H-P filter over the period from 1960 to 2010. The results suggests that there are two distinct cycles in the OECD countries: the Euro area cycle, which includes the business cycles of Germany, France, Italy, Netherlands, Australia and Belgium and the world cycle, which consist of the business cycles of the US, Canada and the U.K.

For a long time, the prevailing view was that we need different theories to explain the different features of the observed data – growth theory to explain the long run trend in a series and something else to explain the cyclical movements

of data. Modern business cycle theory is based on a different premise – the same theory should be capable of explaining both the features. Researchers typically follow one of the two procedures to represent the cyclical component in the data. Some assume a stochastic trend and first difference the (logarithms of the) data to remove it. Most of the authors use a technique known as the Hodrick-Prescott filter (the H-P filter) for representing growth and business cycle components.

Check Your Progress 2

- 1) Give a brief account of Keynesian view on business cycle.

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- 2) Derive the equilibrium income through the model of interaction between multiplier and accelerator.

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- 3) What is meant by conformity and timing of economic series?

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- 4) What is the importance of a lagging indicator?

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11.8 LET US SUM UP

In this unit we focused on three issues, viz., basic features of business cycles , traditional theories of business cycles and measurement of business cycles. Business cycle should be thought of as apparent deviations from a trend. Certain economic variables move together with aggregate output. An examination of the time path of output for any modern industrialized economy reveals that output tends to fluctuate about a long term growth path. These fluctuations about trend

are what we most often think of as the business cycle. The fluctuations are typically irregularly spaced and of varying amplitude and duration. Nevertheless, the one very regular feature of these fluctuations is the way variables move together. Business cycle is characterised by four phases, viz., expansion, recession, depression and recovery. Some of the important mathematical theories which explain these feature of business cycles are: a) Keynesian theory which showed that changes in the level of aggregate effective demand bring about fluctuations in the level of income, output and employment; b) Samuelson model of business cycle which convincingly proved that it is the interaction between the multiplier and accelerator that gives rise to cyclical fluctuations in economic activity, and c) Political theory of business cycle which suggests that business cycles are an outcome of the periodicity of elections in a democracy that prompts the governments to increase public investment before elections and cut down on it afterwards. NBER uses the working definition of business cycle formulated by Burns and Mitchell in the 1940s to identify and date the US business cycle. These dates are widely accepted by government, academic and business analysts. The main problem encountered in the research of business cycle is the lack of a single sufficiently long and consistent measure of aggregate economic activity. The NBER, therefore, relies on the evidence from a large number of comprehensive indicators rather than any single one to date a business cycle. The variables include quantities and prices, stocks and flows, outputs and inputs, real monetary and financial variables. Timing and amplitude of these variables are used to group them into leading, lagging and coincident indicators.

To conclude, a business cycle is not just a movement along a curve in a textbook. It implies new jobs for people, or the loss of them. It is new income, or the loss of it. It is the funds to build new infrastructure such as schools, hospitals and roads. The story of the business cycle is the story of progress and plenty of failure and sacrifice. The effects of recessions extend beyond the purely economic realm and influence the social fabric of society as well. Misery of households, poverty, unemployment, property related crimes, etc. tend to rise during recessions.

11.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Business cycle consists of recurrent alternating phases of expansion and contraction in a large number of economic activities.
- 2) Refer to Sub-Section 11.2 for the important features.
- 3) Describe the phases given at Section 11.3. You should explain Fig. 11.1 also.

Check Your Progress 2

- 1) Keynes emphasized on fluctuation in aggregate demand. Investment multiplier amplified the magnitude of change in output. Refer to Sub-Section 11.4.1.

- 2) Explain the Fig. 11.2. Derive the equilibrium income through the equations (11.1), (11.2) and (11.3) to reach at the equation (11.4). Refer to Sub-Section 11.4.2.
- 3) Conformity is the consistency with which the series have conformed to business cycles. Timing refers to the consistency with which a time series' turning point has led, lagged or roughly coincided with the reference data.
- 4) Refer to Sub-Section 11.6.2. The importance of a lagging indicator is its ability to confirm that a pattern is occurring. Unemployment is one of the most popular lagging indicators. If the unemployment rate is rising, it indicates that the economy has been doing poorly.

